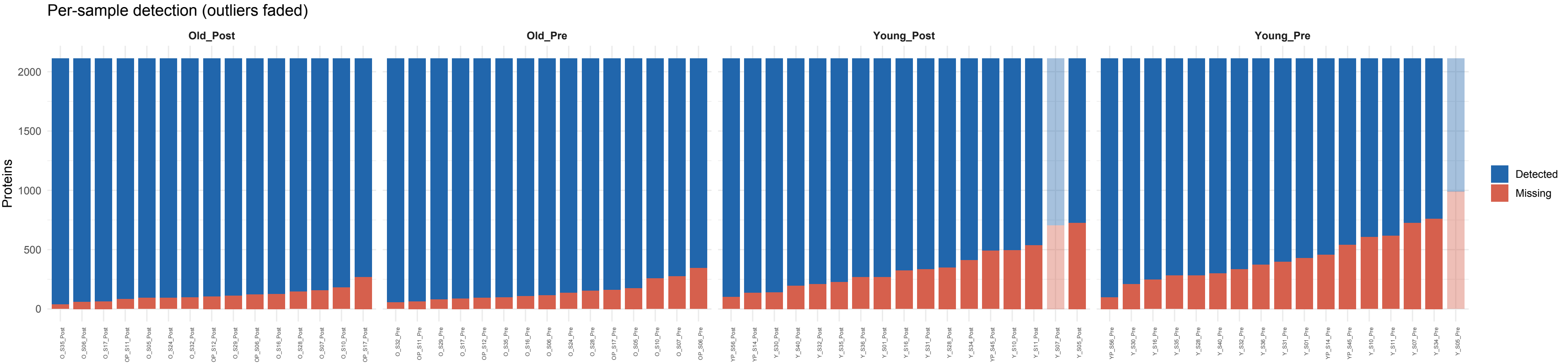
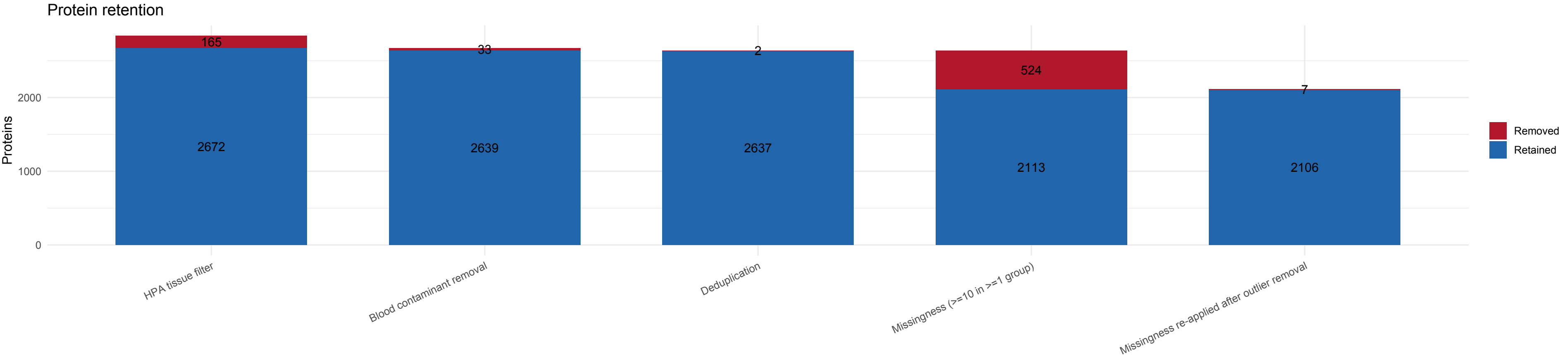


Protein Filtering & Detection

2837 raw → 2106 retained | 62 samples

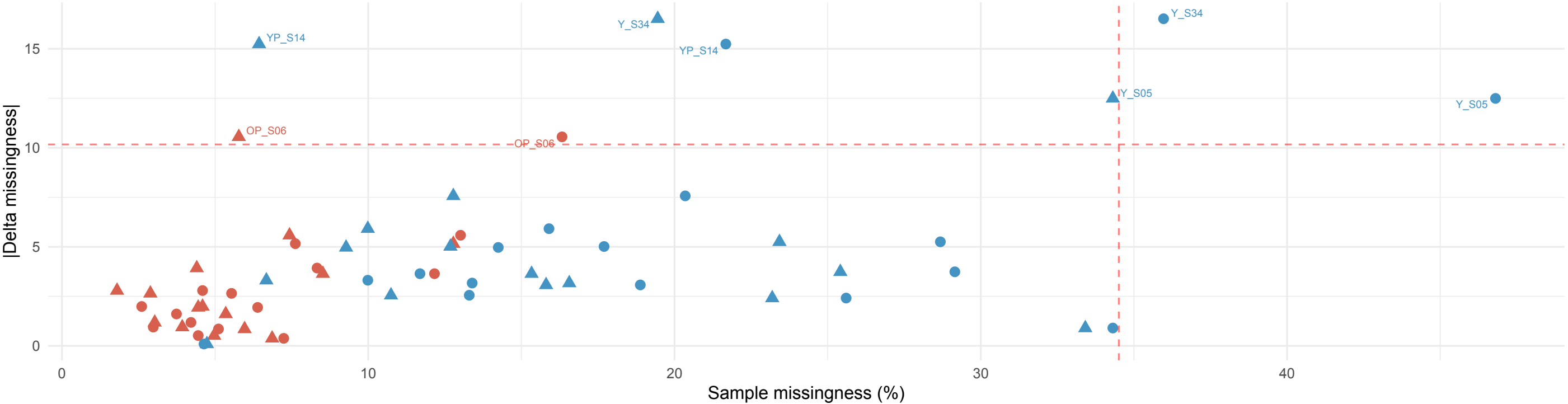


Outlier Diagnostics (4-method consensus)

≥3/4 agreement | 2 removed

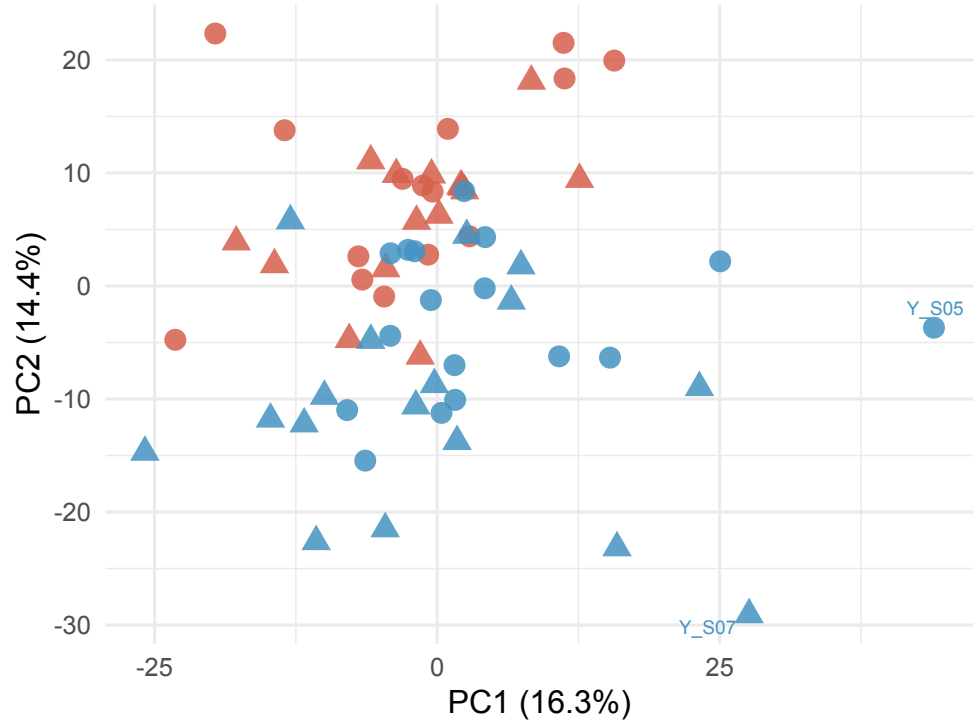
A: Missingness

IQR thresholds: 34.5% / 10.2% | 8 flagged



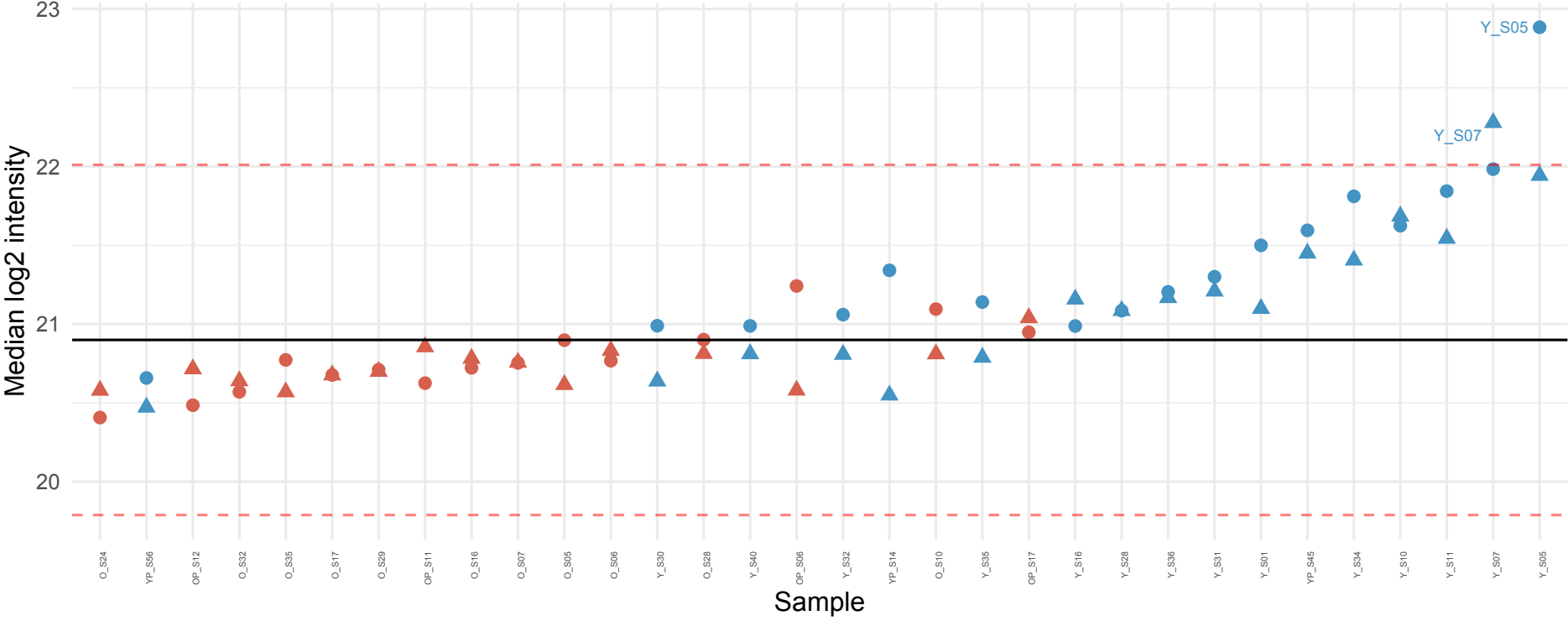
B: PCA Mahalanobis

p < 0.01 | 2 flagged



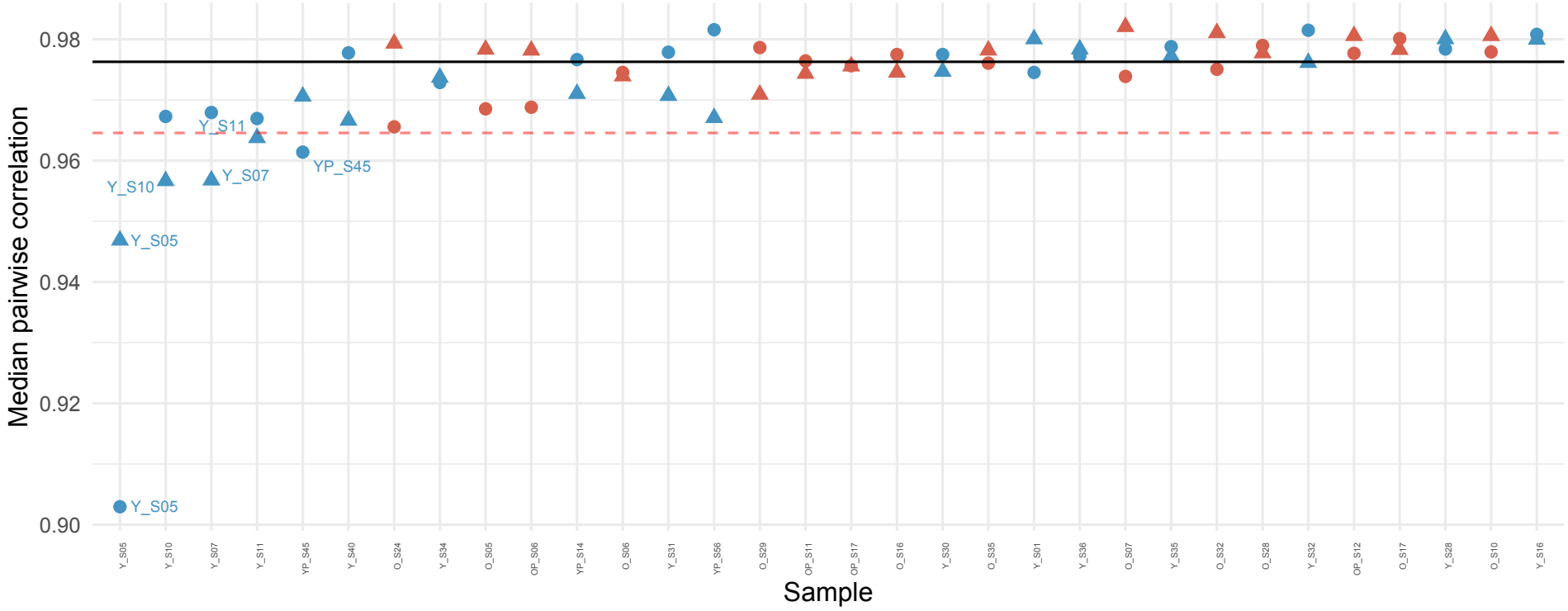
C: MAD median intensity

3x MAD | 2 flagged



D: Inter-sample correlation

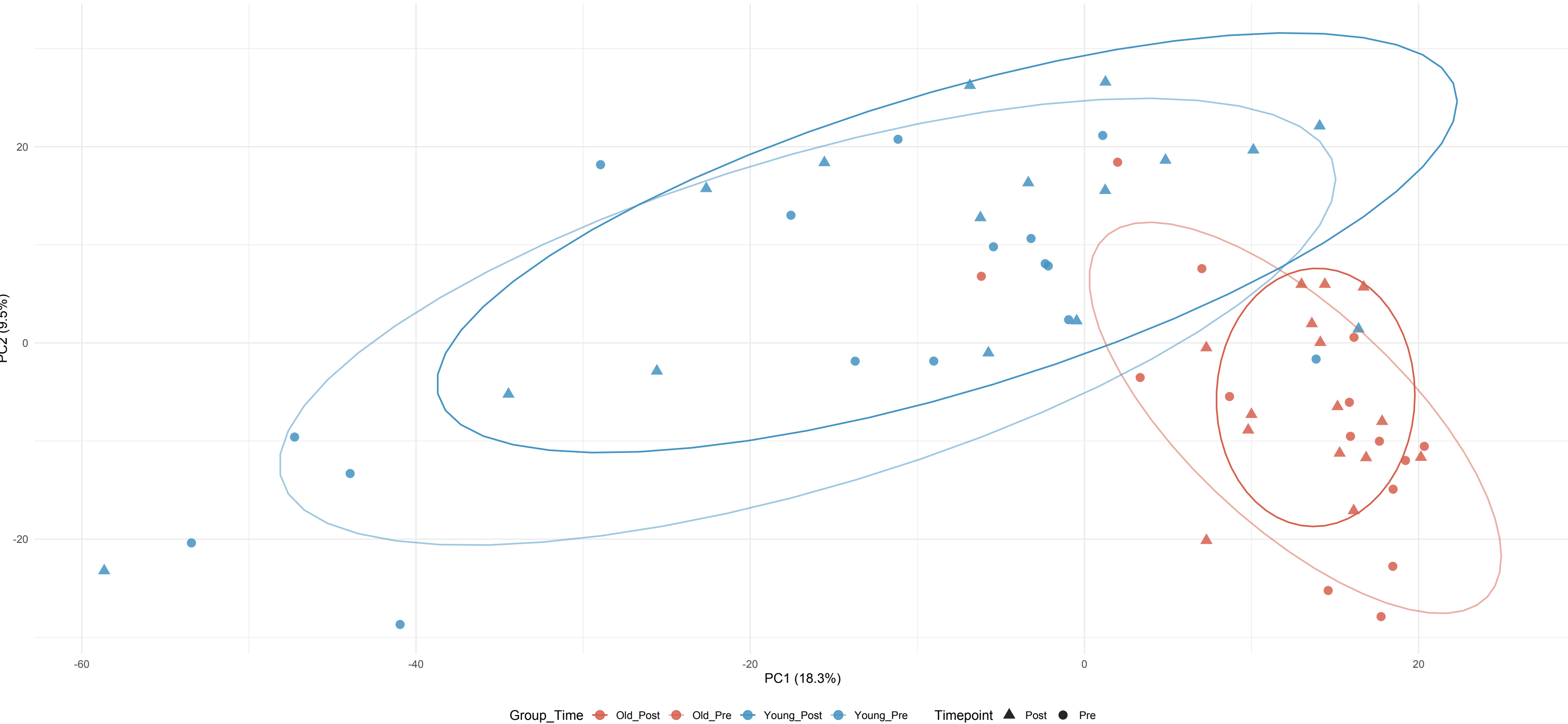
3x MAD | 6 flagged



Timepoint ▲ Post ● Pre age ● Old ● Young

Post-Normalization QC

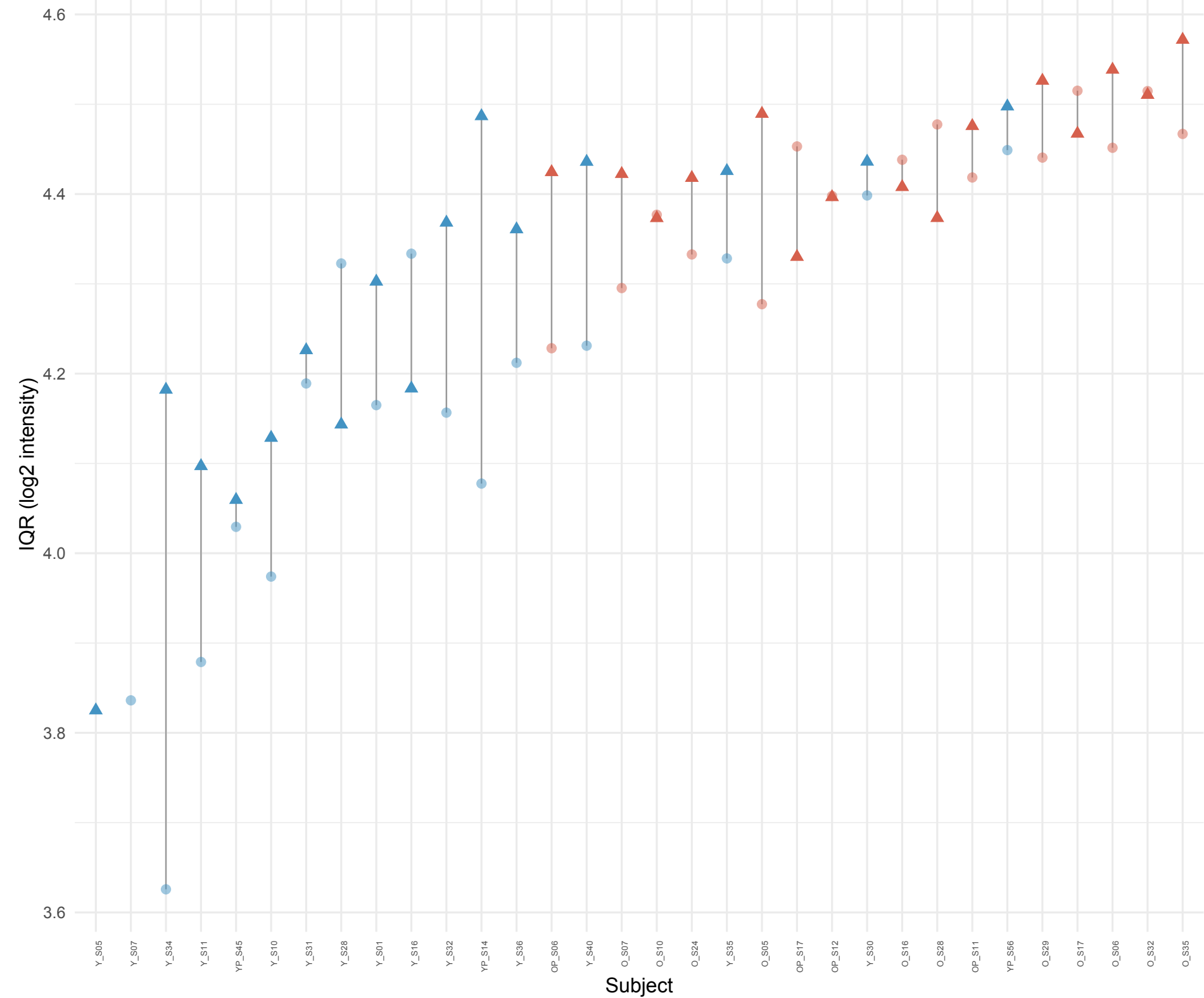
Post-normalization PCA



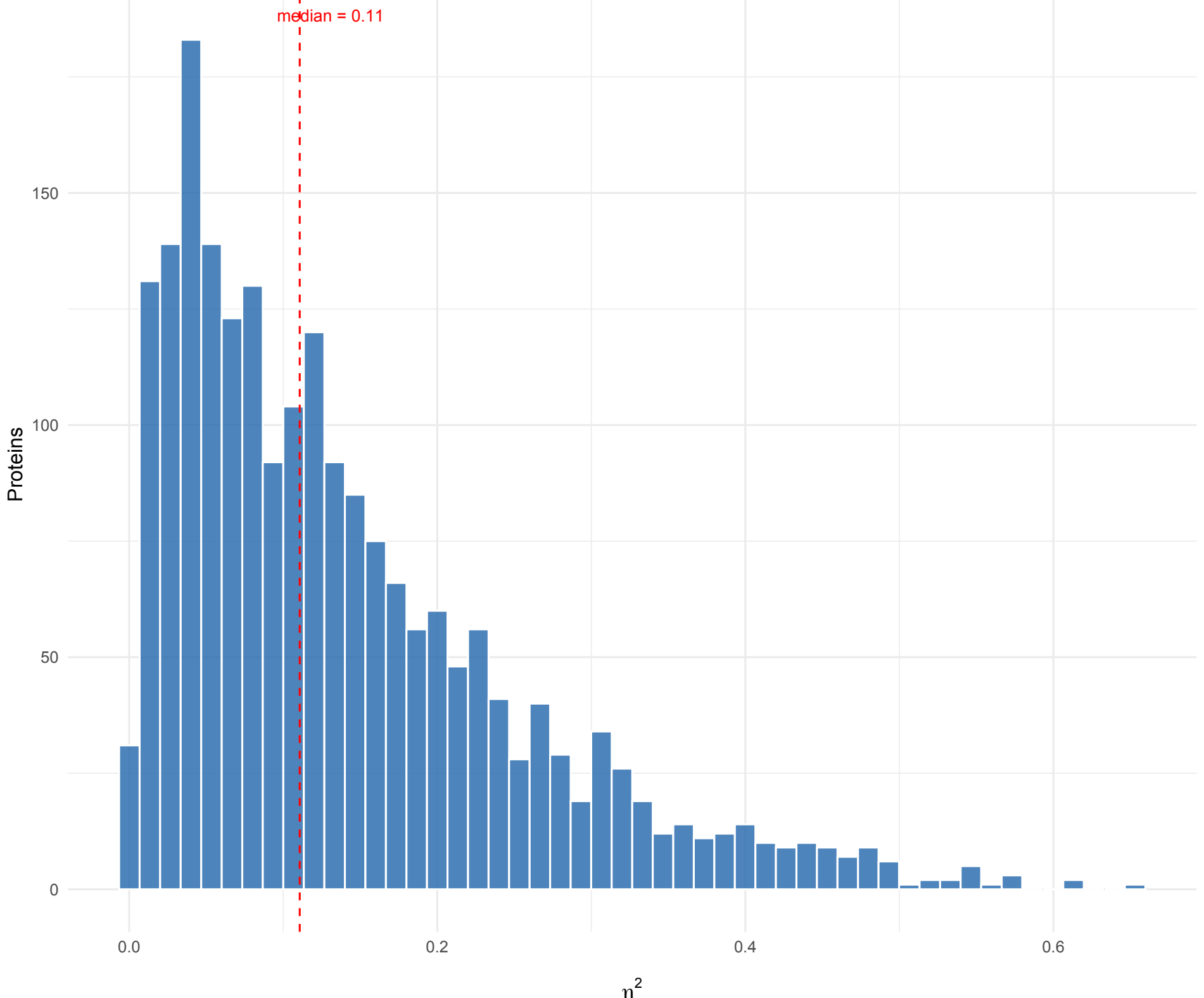
Variability Summary

Per-subject variability

32 subjects | lines connect Pre-Post pairs



Variance partition by group



Group_Time ● Old_Post ● Old_Pre ● Young_Post ● Young_Pre Timepoint ▲ Post ● Pre